# Block practice · More pathogens, immunity and public health

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Explain bacterial growth conditions and how food preservation slows it

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Practice workbook · Explain bacterial growth conditions and how food preservation slows it

Growth-curve phases, temperature groups of bacteria, warmth and moisture, and how chilling, freezing, heating or packaging stop microbes spoiling food.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-MICROBE-GROWTH-AND-FOOD-PRESERVATION

# 1. Question 2 · part b · 6 marks

June 2024 Paper 1

Worked-example retry · Attempt part b only.

**2** (a) The table gives information about different methods of food preservation.

It also gives information on the length of time in days that each method keeps foods fresh.

| Method of food preservation | Time food stays fresh in days |
|---|---|
| refrigeration | 5 |
| freezing | 90 |
| pasteurisation | 35 |
| vacuum-packed food | 14 |

(i) Draw a bar chart to show the information in the table.

**(4)**

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# 1. Question 2 · part b · 6 marks

June 2024 Paper 1

Worked-example retry · Attempt part b only.

(ii) State how many of the methods shown in the bar chart use heat treatment to preserve food.

**(1)**

.....

(b) Bacteria can multiply rapidly in food that is not preserved.

Use words from the box to complete the passage about the growth of bacteria.

**(6)**

| | | | | |
|---|---|---|---|---|
| exponential | more than | stationary | equal to | lag |
| dying | death | less than | live | growth |

The first stage in the growth of bacterial colonies is known as the ..... phase. During this phase, bacteria are not dividing.

The second stage of bacterial growth is known as the ..... phase. During this phase the bacteria are multiplying rapidly.

In the third stage, nutrients are running out and the number of dying cells is ..... the number of dividing cells.

Finally, during the ..... phase the number of dividing cells is ..... the number of ..... cells.

**(Total for Question 2 = 11 marks)**

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## 1. Question 2 · part b · 6 marks · Working and answer

Working and answer

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## 2. Question 4 · part a.ii · 2 marks

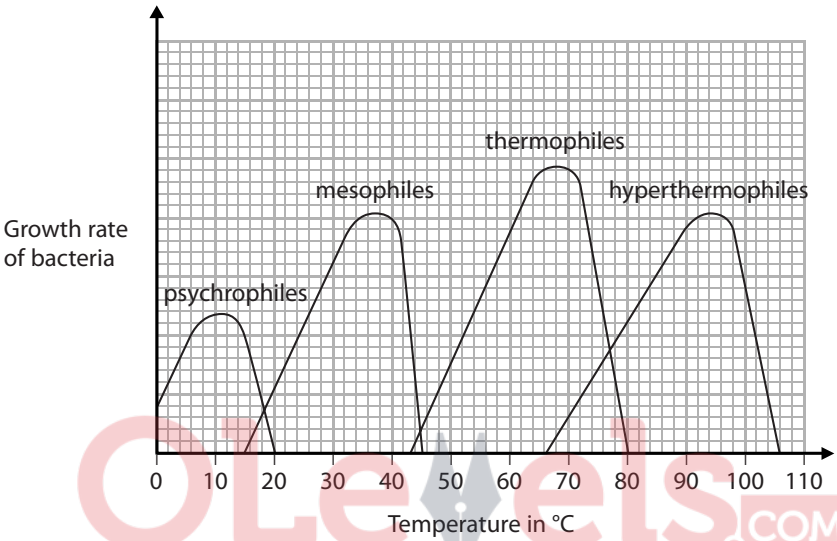
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Further practice · Attempt part a.ii only.

**4** (a) Some bacteria in food can affect the digestive system and cause poor health.

Storing food at the correct temperature can reduce the growth of these bacteria.

The graph shows the growth rate of four groups of bacteria at different temperatures.




(i) Food kept in a fridge shows signs of contamination by bacteria.

Which group of bacteria is most likely to have caused this contamination?

**(1)**

- ☒ **A** psychrophiles
- ☒ **B** mesophiles
- ☒ **C** thermophiles
- ☒ **D** hyperthermophiles

(ii) Explain how cooking food at 65 °C will affect thermophiles.

**(2)**

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## 2. Question 4 · part a.ii · 2 marks

June 2019 Paper 1

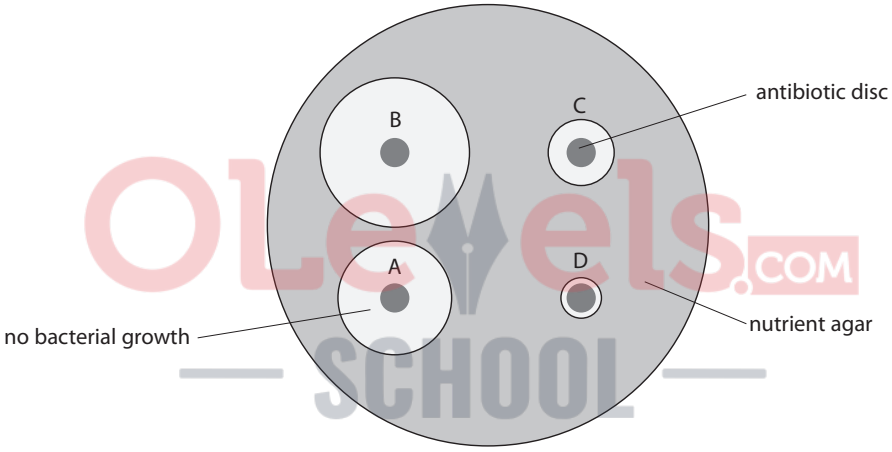
Further practice · Attempt part a.ii only.

(b) A student investigates the effect of different antibiotics on the growth of bacteria.

She uses this method.

- streak bacteria onto nutrient agar in a Petri dish
- place discs of different antibiotics, A, B, C and D, onto the nutrient agar
- incubate the Petri dish in a warm oven for one week

The diagram shows the results after one week.




(i) Explain how the student could determine the effectiveness of each antibiotic.

**(2)**

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## 2. Question 4 · part a.ii · 2 marks · Working and answer

Working and answer

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## 3. Question 5 · part b.ii · 2 marks

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Further practice · Attempt part b.ii only.

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**5** (a) A student reads that as people get older their ability to hear high frequency sounds decreases.

(i) Describe an investigation a student could do to show if this is correct.

**(3)**

(ii) Give a reason why hearing loss occurs as people get older.

**(1)**

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# 3. Question 5 · part b.ii · 2 marks

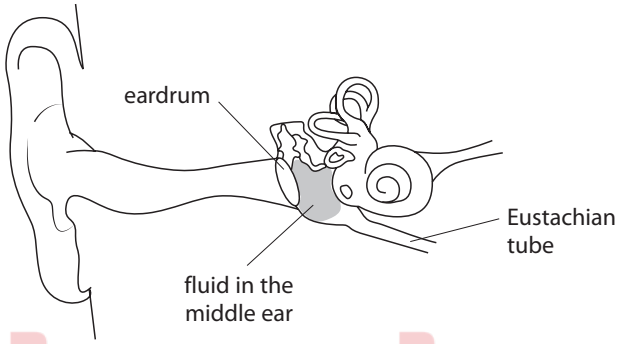
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Further practice · Attempt part b.ii only.

(b) Glue ear is a condition that can affect children and causes hearing loss.

Fluid accumulates in the middle ear, as shown in the diagram.

As the child grows, the slope of the Eustachian tube increases and the fluid drains.




(i) Explain how the fluid in the middle ear causes hearing loss. **(3)**

(ii) Explain why children with glue ear get more bacterial ear infections than other children. **(2)**

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## 3. Question 5 · part b.ii · 2 marks

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Further practice · Attempt part b.ii only.

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(iii) Describe the role of the ear in maintaining a person's balance.

**(3)**

**(Total for Question 5 = 12 marks)**

Working and answer

# 4. Question 4 · part a.i · 1 mark

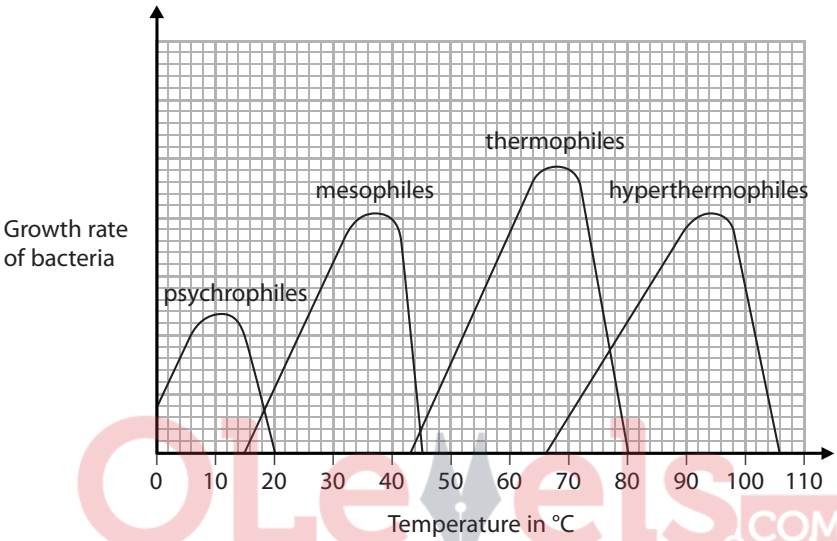
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Further practice · Attempt part a.i only.

**4** (a) Some bacteria in food can affect the digestive system and cause poor health.

Storing food at the correct temperature can reduce the growth of these bacteria.

The graph shows the growth rate of four groups of bacteria at different temperatures.




(i) Food kept in a fridge shows signs of contamination by bacteria.

Which group of bacteria is most likely to have caused this contamination?

**(1)**

- ☒ **A** psychrophiles
- ☒ **B** mesophiles
- ☒ **C** thermophiles
- ☒ **D** hyperthermophiles

(ii) Explain how cooking food at 65 °C will affect thermophiles.

**(2)**

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## 4. Question 4 · part a.i · 1 mark · Working and answer

Working and answer

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## 5. Question 1 · part a · 1 mark

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Further practice · Attempt part a only.

**1** (a) Food packaging gives information about the food it contains.

The photograph shows some of the information given on a package containing yogurt.




(Source: © studiomode / Alamy Stock Photo)

The instructions state that the yogurt should be stored in a fridge at 4°C.

Why should this instruction be followed?

**(1)**

- ☒ **A** to make the yogurt taste better
- ☒ **B** to kill all bacteria in the yogurt
- ☒ **C** to slow down the growth of bacteria in the yogurt
- ☒ **D** to improve the appearance of the yogurt

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# 5. Question 1 · part a · 1 mark

January 2020 Paper 1

Further practice · Attempt part a only.

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(b) The diagram shows the nutritional label on a package containing yogurt.

**Nutrition Facts**

Serving Size
1 container (150g)
Energy in kJ 480
Energy from Fat 0

| Amount/Serving | % Daily Value* | Amount/Serving | % Daily Value* |
|---|---|---|---|
| Total Fat | a% | Sodium | 2% |
| Saturated Fat | 0% | Total Carb. | 6% |
| Trans Fat | | Sugars | |
| Cholesterol | 1% | Protein | 22% |

Vitamin A 4% • Calcium 10% • Vitamin D 20%

Not a significant source of dietary fibre, vitamin C and iron.

(i) State which nutrient on the label is required for growth and repair of body tissue.

**(1)**

.....

(ii) Why is this yogurt suitable for a person who wants to reduce their body mass?

**(1)**

- ☒ **A** it contains vitamins C and D
- ☒ **B** it contains less than 5 mg of cholesterol
- ☒ **C** it does not contain energy from carbohydrates
- ☒ **D** it does not contain energy from fat

(c) Yogurt contains reducing sugars.

Describe a method to test for reducing sugars.

**(3)**

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**(Total for Question 1 = 6 marks)**

## 5. Question 1 · part a · 1 mark · Working and answer

Working and answer

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# 6. Question 6 · part a.iv · 2 marks

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Further practice · Attempt part a.iv only.

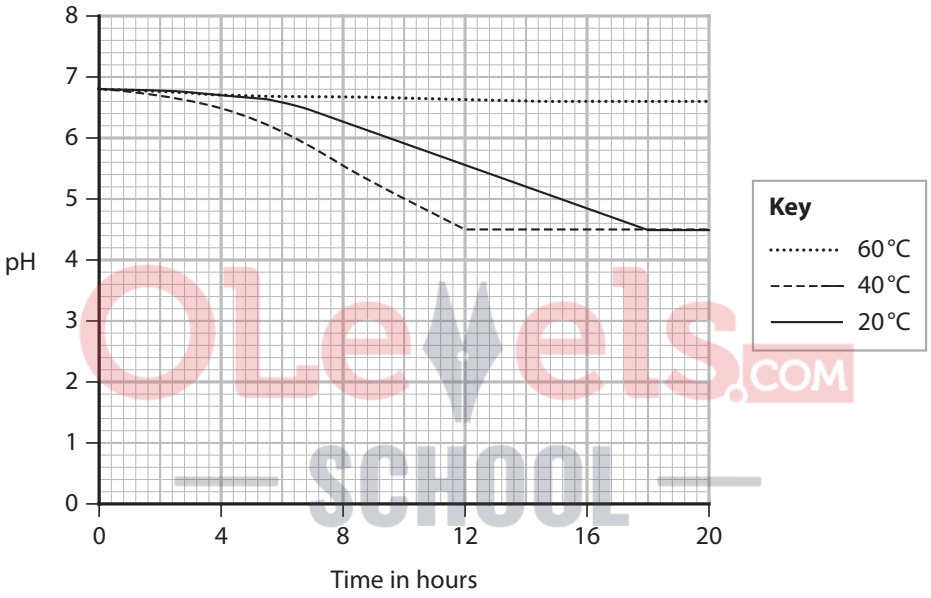
**6** Yoghurt can be produced by incubating milk with a small amount of live yoghurt.

The live yoghurt contains *Lactobacillus* bacteria.

Enzymes convert lactose in the milk into lactic acid causing the pH to decrease to around pH 4.4.

In an investigation, the change in pH as yoghurt is produced was measured during 20 hours at three different temperatures.

The graph shows the results.




(a) (i) Calculate the difference in the time taken to produce yoghurt at 40 °C instead of 20 °C.

**(2)**

time = ..... hours

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# 6. Question 6 · part a.iv · 2 marks

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Further practice · Attempt part a.iv only.

(ii) Give two variables that need to be controlled in this investigation.

**(2)**

1 .....

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2 .....

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(iii) Describe how this investigation could be improved to find the optimum temperature for the production of yoghurt.

**(2)**

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(iv) In some methods to produce yoghurt, the milk is boiled and then cooled before the bacteria are added.

Give the reason for boiling and the reason for cooling the milk.

**(2)**

boiling

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cooling

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# 6. Question 6 · part a.iv · 2 marks

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Further practice · Attempt part a.iv only.

(b) When yoghurt reaches a pH of 4.4 the bacteria stop fermenting lactose.

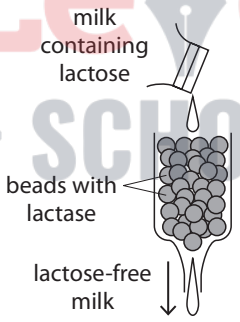
Explain why bacteria stop fermenting the lactose at a pH of around 4.4

**(2)**

(c) People who are lactose intolerant can drink lactose-free milk.

This can be produced using alginate beads containing lactase.

The diagram shows the equipment needed.




(i) Describe the advantages of using alginate beads containing lactase rather than a solution of lactase.

**(2)**

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## 6. Question 6 · part a.iv · 2 marks · Working and answer

Working and answer

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# 7. Question 2 · part a.ii · 1 mark

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Further practice · Attempt part a.ii only.

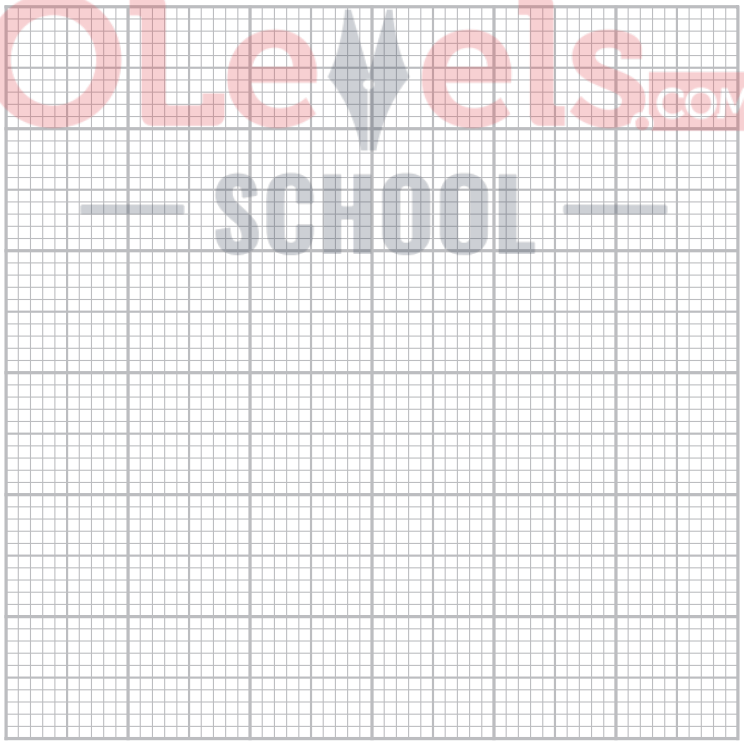
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(i) Draw a bar chart to show the information in the table.

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# 7. Question 2 · part a.ii · 1 mark

June 2024 Paper 1

Further practice · Attempt part a.ii only.

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(ii) State how many of the methods shown in the bar chart use heat treatment to preserve food.

**(1)**

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(b) Bacteria can multiply rapidly in food that is not preserved.

Use words from the box to complete the passage about the growth of bacteria.

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**(Total for Question 2 = 11 marks)**

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## 7. Question 2 · part a.ii · 1 mark · Working and answer

Working and answer

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